



Chapter: Motion in a Plane

Prep for RT-03 · Sun 16 Aug 2026

Take $g = 10 \text{ m/s}^2$ unless stated

24 Questions · 4 Sections + Answer Key

CONCEPT RECAP

$$\vec{v} = \vec{u} + \vec{a}t$$

$$v_x = u_x + a_x t$$

$$v_y = u_y + a_y t$$

- Valid **only** when acceleration $\vec{a}$ is constant in **both magnitude and direction** over the interval.
- In a plane, treat x and y as two **independent 1-D motions** — solve each axis separately, then combine as vectors.
- Commonest CET use: **projectile motion** → $a_x = 0$, $a_y = -g$ (taking up as $+y$). Horizontal velocity never changes.
- ⚠ NOT directly valid over a finite time interval in **uniform circular motion** — $|a|$ is constant but its *direction* keeps rotating, so $\vec{a}t$ cannot be added as one fixed vector.

WORKED EXAMPLE — METHOD TO FOLLOW

Q. A particle is projected with $u = 10 \text{ m/s}$ at 30° above the horizontal. Find its velocity vector at $t = 1 \text{ s}$. ($g = 10 \text{ m/s}^2$, $\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.866$)

- Split into components: $u_x = 10 \times 0.866 = 8.66 \text{ m/s}$, $u_y = 10 \times 0.5 = 5 \text{ m/s}$
- Apply $v_x = u_x + a_x t \rightarrow a_x = 0$, so $v_x = 8.66 \text{ m/s}$ (unchanged)
- Apply $v_y = u_y + a_y t \rightarrow v_y = 5 + (-10)(1) = -5 \text{ m/s}$
- Recombine: $\vec{v} = 8.66\hat{i} - 5\hat{j} \text{ m/s} \rightarrow \text{speed} = \sqrt{(8.66^2 + 5^2)} = \sqrt{100} = \mathbf{10 \text{ m/s}}$, directed 30° below horizontal

Section A · Warm-Up (Straight-Line Recap)

Rebuild the base equation before moving to 2-D. All motion here is along one line.

A1. A car starts from rest and accelerates uniformly at 2 m/s^2 . Its velocity after 5 s is:

- | | |
|-----------|-----------|
| a) 5 m/s | b) 10 m/s |
| c) 15 m/s | d) 20 m/s |

A2. A particle moving at 20 m/s decelerates at 4 m/s^2 . Time taken to come to rest:

- | | |
|--------|--------|
| a) 4 s | b) 5 s |
| c) 6 s | d) 8 s |

A3. A body has $u = 15 \text{ m/s}$ and $a = -3 \text{ m/s}^2$. Its velocity after 4 s is:

- | | |
|----------|-----------|
| a) 3 m/s | b) 6 m/s |
| c) 9 m/s | d) 12 m/s |

A4. The correct vector form of the first equation of motion is:

- | | |
|----------------------|-----------------------------------|
| a) $v = u + at$ | b) $\vec{v} = \vec{u} + \vec{a}t$ |
| c) $v^2 = u^2 + 2as$ | d) $s = ut + \frac{1}{2}at^2$ |

A5. A particle has $u = 10 \text{ m/s}$, $a = 2 \text{ m/s}^2$ for the first 3 s, then $a = 0$ for the next 2 s. Its velocity at $t = 5 \text{ s}$ is:

- | | |
|-----------|-----------|
| a) 16 m/s | b) 20 m/s |
| c) 22 m/s | d) 26 m/s |

Section B · Projectile Motion (Component Method)

$a_x = 0$, $a_y = -g$. Questions B3–B8 share the same launch — do them in order.

B1. A ball is projected horizontally at 20 m/s off a cliff. Its vertical velocity component after 2 s is:

- a) 10 m/s
- b) 20 m/s
- c) 30 m/s
- d) 40 m/s

B2. For the same ball (B1), its resultant speed at $t = 2$ s is:

- a) 20 m/s
- b) 24 m/s
- c) 28.3 m/s
- d) 40 m/s

B3. A projectile is launched at 30 m/s at 37° above horizontal ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$). Its horizontal and vertical velocity components (u_x , u_y) are:

- a) 24, 18 m/s
- b) 18, 24 m/s
- c) 15, 26 m/s
- d) 20, 22 m/s

B4. For the same projectile (B3), the vertical velocity component at $t = 1$ s is:

- a) 8 m/s
- b) 10 m/s
- c) 18 m/s
- d) 28 m/s

B5. For the same projectile (B3), the time to reach maximum height ($v_y = 0$) is:

- a) 1.2 s
- b) 1.5 s
- c) 1.8 s
- d) 2.0 s

B6. For the same projectile (B3), the magnitude of vertical velocity at $t = 3$ s is:

- a) 8 m/s
- b) 10 m/s
- c) 12 m/s
- d) 14 m/s

B7. A stone is thrown at 45° with $u_x = u_y = 20$ m/s. Its vertical velocity at $t = 1$ s is:

- a) 0 m/s
- b) 10 m/s
- c) 15 m/s
- d) 20 m/s

B8. For the same stone (B7), its speed at $t = 1$ s is:

- a) 20 m/s
- b) 22.4 m/s
- c) 25 m/s
- d) 30 m/s

Section C · General 2-D Motion & Circular Motion (Concept Check)

C1–C2 use vector algebra directly. C3–C4 test WHERE the equation does and doesn't apply.

C1. A particle has $\vec{u} = (4\hat{i} - 1\hat{j})$ m/s and constant $\vec{a} = (2\hat{i} + 3\hat{j})$ m/s². Its velocity at $t = 2$ s is:

- a) $8\hat{i} + 5\hat{j}$
- b) $4\hat{i} + 3\hat{j}$
- c) $6\hat{i} + 2\hat{j}$
- d) $8\hat{i} + 6\hat{j}$

C2. The speed of the particle in C1 at $t = 2$ s is closest to:

- a) 7.8 m/s
- b) 8.5 m/s
- c) 9.4 m/s
- d) 10.2 m/s

C3. $\vec{v} = \vec{u} + \vec{a}t$ applies directly to which of these?

- a) Uniform circular motion over 5 s
- b) Projectile motion under gravity
- c) A car turning a corner at constant speed
- d) Motion where $\vec{a}$'s direction keeps changing

C4. A cyclist moves at constant speed 10 m/s around a circular track. Applying $\vec{v} = \vec{u} + \vec{a}t$ over a 3 s interval is:

- a) Valid — speed doesn't change
- b) Invalid — centripetal acceleration's direction keeps changing
- c) Valid — $|a|$ is constant
- d) Valid only for large radius

C5. A particle has $\vec{u} = (6\hat{i} + 8\hat{j})$ m/s and constant $\vec{a} = (-4\hat{j})$ m/s². The time at which its y-component of velocity becomes zero is:

- a) 1 s
- b) 1.5 s
- c) 2 s
- d) 2.5 s

Section D · CET-Level Mixed (Previous-Year Style)

Multi-step. These are the ones that separate 70% from 90%+.

D1. Two projectiles are launched with the same speed u — one at angle θ , the other at $(90^\circ - \theta)$ — landing at the same range R . If t_1, t_2 are their times of flight, the correct relation is:

- a) $t_1 t_2 = 2R/g$
- b) $t_1 t_2 = R/g$
- c) $t_1 + t_2 = R/g$
- d) $t_1 - t_2 = 2u/g$

D2. A ball is projected at 60° with $u = 20$ m/s ($\sin 60^\circ = 0.866$). The time at which its vertical velocity drops to exactly half of u_y is:

- a) 0.5 s
- b) 0.87 s
- c) 1.2 s
- d) 1.7 s

D3. A particle's velocity changes from $(3\hat{i} + 4\hat{j})$ m/s to $(7\hat{i} + 4\hat{j})$ m/s in 2 s under constant acceleration. That acceleration is:

- a) $2\hat{i}$ m/s²
- b) $4\hat{i}$ m/s²
- c) $2\hat{i} + 2\hat{j}$ m/s²
- d) $1\hat{i}$ m/s²

D4. If the same acceleration (D3) continues from the same start, the velocity at $t = 5$ s is:

- a) $13\hat{i} + 4\hat{j}$
- b) $10\hat{i} + 4\hat{j}$
- c) $13\hat{i}$
- d) $8\hat{i} + 4\hat{j}$

D5. A body starts at $\vec{u} = 5\hat{i}$ m/s with constant $\vec{a} = -10\hat{j}$ m/s². The time at which its velocity makes 45° with the horizontal is:

- a) 0.25 s
- b) 0.5 s
- c) 0.75 s
- d) 1.0 s

D6. For a projectile (air resistance neglected), splitting $\vec{v} = \vec{u} + \vec{a}t$ into $v_x = u_x$ and $v_y = u_y - gt$ is:

- a) Valid — x and y motions are independent when a is constant along one axis
- b) Invalid — components can never be separated
- c) Valid only at the highest point
- d) Invalid — horizontal velocity also decreases

ANSWER KEY + ONE-LINE WORKING

Q	Working
A1	b $v = 0 + 2 \times 5 = 10 \text{ m/s}$
A2	b $0 = 20 - 4t \rightarrow t = 5 \text{ s}$
A3	a $v = 15 - 3 \times 4 = 3 \text{ m/s}$
A4	b Vector quantities need vector notation: $\vec{v} = \vec{u} + \vec{a}t$
A5	a v after 3s = $10 + 2 \times 3 = 16 \text{ m/s}$; $a = 0$ for next 2s so v stays 16 m/s
B1	b $v_y = 0 + 10 \times 2 = 20 \text{ m/s}$
B2	c $v = \sqrt{(20^2 + 20^2)} = \sqrt{800} \approx 28.3 \text{ m/s}$
B3	a $u_x = 30 \times 0.8 = 24$, $u_y = 30 \times 0.6 = 18 \text{ m/s}$
B4	a $v_y = 18 - 10 \times 1 = 8 \text{ m/s}$
B5	c $t = u_y/g = 18/10 = 1.8 \text{ s}$
B6	c $v_y = 18 - 10 \times 3 = -12 \text{ m/s} \rightarrow$ magnitude 12 m/s
B7	b $v_y = 20 - 10 \times 1 = 10 \text{ m/s}$
B8	b $v = \sqrt{(20^2 + 10^2)} = \sqrt{500} \approx 22.4 \text{ m/s}$
C1	a $v_x = 4 + 2 \times 2 = 8$, $v_y = -1 + 3 \times 2 = 5 \rightarrow 8\hat{i} + 5\hat{j}$
C2	c $ v = \sqrt{(8^2 + 5^2)} = \sqrt{89} \approx 9.4 \text{ m/s}$
C3	b Gravity gives constant $\vec{a}$ in both magnitude and direction throughout flight
C4	b Centripetal $\vec{a}$ keeps rotating — can't be added as one fixed vector over the interval
C5	c $0 = 8 - 4t \rightarrow t = 2 \text{ s}$
D1	a $t_1 t_2 = (2u \sin\theta/g)(2u \cos\theta/g) = 2u^2 \sin 2\theta/g^2 = 2R/g$
D2	b $u_y = 20 \times 0.866 = 17.32$; half = 8.66; $8.66 = 17.32 - 10t \rightarrow t = 0.87 \text{ s}$
D3	a $\vec{a} = (\vec{v} - \vec{u})/t = (4\hat{i} + 0\hat{j})/2 = 2\hat{i} \text{ m/s}^2$
D4	a $v_x = 3 + 2 \times 5 = 13$, $v_y = 4$ ($a_y = 0$) $\rightarrow 13\hat{i} + 4\hat{j}$
D5	b 45° needs $ v_y = v_x = 5 \rightarrow 10t = 5 \rightarrow t = 0.5 \text{ s}$
D6	a Perpendicular components of constant acceleration act independently